MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

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Review Questions | Part 1 (with Solutions)

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These are review questions for Chapters 1, 2, 3 in Ross' book.

(1) A president, treasurer, and secretary, all different, are to be chosen from a club consisting of 10 people. How many different choices of officers are possible if
(a) there are no restrictions?
(b) A and B will not serve together?
(c) C and D will serve together or not at all?
(d) E must be an officer?
(e) F will serve only if he is president?

Solution:
(a) $10 \cdot 9 \cdot 8 = 720$.
(b) $8 \cdot 7 \cdot 6 + 2 \cdot 3 \cdot 8 \cdot 7 = 672$. Here is the idea: there are $8 \cdot 7 \cdot 6$ choices not including A or B and there are $3 \cdot 8 \cdot 7$ choices in which a specified one of A and B, but not the other, serves.
(c) $8 \cdot 7 \cdot 6 + 3 \cdot 2 \cdot 8 = 384$.
(d) $3 \cdot 9 \cdot 8 = 216$.
(e) $9 \cdot 8 \cdot 7 + 9 \cdot 8 = 576$.

(2) How many different 7-place license plates are possible when 3 of the entries are letters and 4 are digits? Assume that repetition of letters and numbers is allowed and that there is no restriction on where the letters or numbers can be placed.

Solution: There are $\binom{7}{3} = 35$ choices of the three places for the letters. For each choice, there are $(26)^3(10)^4$ different license plates. Hence, altogether there are $35 \cdot (26)^3 \cdot (10)^4$ different plates.

(3) Consider three classes, each consisting of n students. From this group of $3n$ students, a group of 3 students is to be chosen.
(a) How many choices are possible?
(b) How many choices are there in which all 3 students are in the same class?
(c) How many choices are there in which 2 of the 3 students are in the same class and the other student is in a different class?
(d) How many choices are there in which all 3 students are in different classes?
(e) Using the results of parts (a) through (d), write a combinatorial identity.

Solution:
(a) $\binom{3n}{3}$.
(b) $3\binom{n}{3}$.
(c) $\binom{3}{1}\binom{2}{1}\binom{n}{2}\binom{n}{1} = 3n^2(n-1)$.
(d) n^3.

(e) $\binom{3n}{n} = 3\binom{n}{3} + 3n^2(n-1) + n^3.$

(4) A committee of 6 people is to be chosen from a group consisting of 7 men and 8 women. If the committee must consist of at least 3 women and at least 2 men, how many different committees are possible?

Solution: $\binom{8}{3}\binom{7}{3} + \binom{8}{4}\binom{7}{2} = 3430.$

(5) An urn A contains 3 red and 3 black balls, whereas urn B contains 4 red and 6 black balls. If a ball is randomly selected from each urn, what is the probability that the balls will be the same color?

Solution: Let R be the event that both balls are red, and let B be the event that both are black. Then

$$P(R \cup B) = P(R) + P(B) = \frac{3 \cdot 4}{6 \cdot 10} + \frac{3 \cdot 6}{6 \cdot 10} = \frac{1}{2}.$$

(6) From a group of 3 freshmen, 4 sophomores, 4 juniors, and 3 seniors a committee of size 4 is randomly selected. Find the probability that the committee will consist of
(a) 1 from each class;
(b) 2 sophomores and 2 juniors;
(c) only sophomores or juniors.

Solution: (a) $\frac{3 \cdot 4 \cdot 4 \cdot 3}{\binom{14}{4}} = 0.1439.$ (b) $\frac{\binom{4}{2}\binom{4}{2}}{\binom{14}{4}} = 0.0360.$ (c) $\frac{\binom{8}{4}}{\binom{14}{4}} = 0.0699.$

(7) Ten cards are randomly chosen from a deck of 52 cards that consists of 13 cards of each of 4 different suits. Each of the selected cards is put in one of 4 piles, depending on the suit of the card.
(a) What is the probability that the largest pile has 4 cards, the next largest has 3, the next largest has 2, and the smallest has 1 card?
(b) What is the probability that two of the piles have 3 cards, one has 4 cards, and one has no cards?

Solution:
(a) The probability that the 10 cards consist of 4 spades, 3 hearts, 2 diamonds, and 1 club is $\frac{\binom{14}{4}\binom{13}{3}\binom{13}{2}\binom{13}{1}}{\binom{52}{10}}$. Because there are 4! possible choices of the suits to have 4, 3, 2, and 1 cards, respectively, it follows that the probability is $\frac{4!\binom{14}{4}\binom{13}{3}\binom{13}{2}\binom{13}{1}}{\binom{52}{10}}$.
(b) Because there are $\binom{4}{2} = 6$ choices of the two suits that are to have 3 cards and then 2 choices for the suit to have 4 cards, the probability is $\frac{12\binom{13}{3}\binom{13}{3}\binom{13}{4}}{\binom{52}{10}}$.

(8) Let $T_k(n)$ denote the number of partitions of the set $\{1, \cdots, n\}$ into k nonempty subsets, where $1 \leq k \leq n$. Prove that

$$T_k(n) = kT_k(n-1) + T_{k-1}(n-1).$$

Hint: In how many partitions is 1 a subset, and in how many is 1 an element of a subset that contains other elements?

Solution: The number of partitions for which $\{1\}$ is a subset is equal to the number of partitions of the remaining $n-1$ elements into $k-1$ nonempty subsets, namely, $T_{k-1}(n-1)$. Because there are $T_k(n-1)$ partitions of $\{2, \cdots, n-1\}$ into k nonempty subsets and then a choice of k of them in which to place element 1, it follows that there are $kT_k(n-1)$ partitions for which $\{1\}$ is not a subset. This completes the proof.

(9) Five balls are randomly chosen, without replacement, from an urn that contains 5 red, 6 white, and 7 blue balls. Find the probability that at least one ball of each color is chosen.

Solution: Let R, W, B denote, respectively, the events that there are NO red, NO white, and NO blue balls chosen. Then

$$\begin{aligned} P(R \cup W \cup B) &= P(R) + P(W) + P(B) - P(RW) - P(RB) - P(WB) + P(RWB) \\ &= \frac{\binom{13}{5}}{\binom{18}{5}} + \frac{\binom{12}{5}}{\binom{11}{5}} + \frac{\binom{11}{5}}{\binom{11}{5}} - \frac{\binom{7}{5}}{\binom{18}{5}} - \frac{\binom{6}{5}}{\binom{18}{5}} - \frac{\binom{5}{5}}{\binom{18}{5}} \approx 0.2933. \end{aligned}$$

This gives the wanted probability $P(R' \cap W' \cap B') \approx 0.7067$.

(10) Balls are randomly removed from an urn initially containing 20 red and 10 blue balls. What is the probability that all of the red balls are removed before all of the blue ones have been removed?

Solution: All the red balls are removed before all the blue ones if and only if the very last ball removed is blue. Because all 30 balls are equally likely to be the last ball removed, the probability is 10/30=1/3.

(11) The probability that a new car battery functions for over 10,000 miles is .8, the probability that it functions for over 20,000 miles is .4, and the probability that it functions for over 30,000 miles is .1. If a new car battery is still working after 10,000 miles, what is the probability that
(a) its total life will exceed 20,000 miles?
(b) its additional life will exceed 20,000 miles?

Solution: Let L_i denote the event that the life of the battery is greater than $10,000 \times i$ miles. Then
(a) $P(L_2|L_1) = P(L_1L_2)/P(L_1) = P(L_2)/P(L_1) = 0.4/0.8 = 1/2$;
(b) $P(L_3|L_1) = P(L_1L_3)/P(L_1) = P(L_3)/P(L_1) = 0.1/0.8 = 1/8$.

(12) An urn A contains 2 white balls and 1 black ball, whereas urn B contains 1 white ball and 5 black balls. A ball is drawn at random from urn A and placed in urn B. A ball is then drawn from urn B. It happens to be white. What is the probability that the ball transferred was white?

Solution: Let T be the event that the transferred ball is white, and let W be the event that a white ball is drawn from urn B. Then

$$\begin{aligned} P(T|W) &= \frac{P(W|T)P(T)}{P(W|T)P(T) + P(W|T')P(T')} \\ &= \frac{(2/7)(2/3)}{(2/7)(2/3) + (1/7)(1/3)} = \frac{4}{5}. \end{aligned}$$

(13) A friend randomly chooses two cards, without replacement, from an ordinary deck of 52 playing cards. In each of the following situations, determine the conditional probability that both cards are aces.

(a) You ask your friend if one of the cards is the ace of spades, and your friend answers in the affirmative.
(b) You ask your friend if the first card selected is an ace, and your friend answers in the affirmative.
(c) You ask your friend if the second card selected is an ace, and your friend answers in the affirmative.
(d) You ask your friend if either of the cards selected is an ace, and your friend answers in the affirmative.

Solution: Let B denote the event that both cards are aces.
(a)

$$\begin{aligned} P(B|\text{yes to ace of spades}) &= \frac{P(B, \text{yes to ace of spades})}{P(\text{yes to ace of spades})} \\ &= \frac{\binom{1}{1}\binom{3}{1}/\binom{52}{2}}{\binom{1}{1}\binom{51}{1}/\binom{52}{2}} = \frac{1}{17}. \end{aligned}$$

(b) Since the second card is equally likely to be any of the remaining 51, of which 3 are aces, we see that the answer in this situation is also 1/17.
(c) Because we can always interchange which card is considered first and which is considered second, the result should be the same as in part (b). A more formal argument is as follows:

$$\begin{aligned} P\{B|\text{second is ace}\} &= \frac{P\{B, \text{second is ace}\}}{P\{\text{second is ace}\}} \\ &= \frac{P(B)}{P(B) + P\{\text{first is not ace, second is ace}\}} \\ &= \frac{(4/52)(3/51)}{(4/52)(3/51) + (48/52)(4/51)} = \frac{1}{17}. \end{aligned}$$

(d)

$$\begin{aligned} P\{B|\text{at least one}\} &= \frac{P\{B\}}{P\{\text{at least one}\}} \\ &= \frac{(4/52)(3/51)}{1 - (48/52)(47/51)} = \frac{1}{33}. \end{aligned}$$

(14) A type C battery is in working condition with probability 0.7, whereas a type D battery is in working condition with probability 0.4. A battery is randomly chosen from a bin consisting of 8 type C and 6 type D batteries.
(a) What is the probability that the battery works?
(b) Given that the battery does not work, what is the conditional probability that it was a type C battery?

Solution: Let W be the event that the battery works, and let C and D denote the events that the battery is a type C and that it is a type D battery, respectively.
(a) $P(W) = P(W|C)P(C) + P(W|D)P(D) = 0.7(8/14) + 0.4(6/14) = 4/7$.
(b) $P(C|W') = \dfrac{P(CW')}{P(W')} = \dfrac{P(W'|C)P(C)}{P(W')} = \dfrac{0.3 \times (8/14)}{3/7} = 0.4$.

(15) Two local factories, A and B, produce radios. Each radio produced at factory A is defective with probability 0.05, whereas each one produced at factory B is defective with probability 0.01. Suppose you purchase two radios that were produced at the same factory, which is equally likely to have been either factory A or factory B. If the first radio that you check is defective, what is the conditional probability that the other one is also defective?

Solution: Let D_i, $i = 1, 2$, denote the event that radio i is defective. Also, let A and B be the events that the radios were produced at factory A and at factory B, respectively. Then

$$\begin{aligned} P(D_2|D_1) &= \frac{P(D_1D_2)}{P(D_1)} \\ &= \frac{P(D_1D_2|A)P(A) + P(D_1D_2|B)P(B)}{P(D_1|A)P(A) + P(D_1|B)P(B)} \\ &= \frac{(0.05)^2(1/2) + (0.01)^2(1/2)}{(0.05)(1/2) + (0.01)(1/2)} \\ &= \frac{13}{300}. \end{aligned}$$